

### ENGINEERING SPECIFICATIONS

Document Name: STANDARD OFFERINGS FOR ELECTRICAL PANEL

|                  |                 |                  |                  |
|------------------|-----------------|------------------|------------------|
| Document No:     | ESP-17-18-101   | Revision: R04    | Date: 14/04/2022 |
| Prepared By: MRM | Revised By: SRS | Approved By: ASP | Sheet: 1 of 2    |

### STANDARD OFFERINGS

| SR.NO  | SERIES              | TYPE OF STARTER                                       | ISOLATION                   | PROTECTION                                | PANELS & MOUNTING PROVISION                  |                                                         |  |
|--------|---------------------|-------------------------------------------------------|-----------------------------|-------------------------------------------|----------------------------------------------|---------------------------------------------------------|--|
|        |                     |                                                       |                             |                                           | CONTROL PANEL                                | STARTER PANEL                                           |  |
| 1      | KSC                 | VFD                                                   | SDU with door handle        | FUSES                                     | UNIT MOUNTED                                 | STANDALONE                                              |  |
|        |                     | SOFT STARTER                                          | SDU with door handle        | FUSES                                     |                                              |                                                         |  |
|        |                     | STAR DELTA (Compulsory with close transition starter) | SDU with door handle        | FUSES                                     |                                              |                                                         |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 2a.    | KWS                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | STAR DELTA                                            | MCCB                        |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFT STARTER                                          | SDU                         | FUSES                                     | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | STAR DELTA                                            | MCCB per circuit            |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFT STARTER                                          | COMMON SDU                  | FUSES per circuit                         | UNIT MOUNTED                                 | UNIT MOUNTED                                            |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 2b.    | KWS (SPECIAL BRINE) |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | PART WINDING                                          | MCCB                        |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | VFD                                                   | SDU                         | FUSES                                     | UNIT MOUNTED                                 | STANDALONE                                              |  |
|        | DOUBLE CKT          | PART WINDING                                          | MCCB per circuit            |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | VFD                                                   | COMMON SDU                  | FUSES per circuit                         | UNIT MOUNTED                                 | STANDALONE                                              |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 3a.    | KWK                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | STAR DELTA                                            | MCCB                        |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFT STARTER                                          | SDU                         | FUSES                                     | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | STAR DELTA                                            | MCCB per circuit            |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFT STARTER                                          | COMMON SDU                  | FUSES per circuit                         | UNIT MOUNTED                                 | UNIT MOUNTED                                            |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 3b.    | KWK -(OPTIONAL)     |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | STAR DELTA                                            | Not Available               | FUSES                                     | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | STAR DELTA                                            |                             | FUSES per circuit                         | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 4      | KWI                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | VFD                                                   | SDU                         | FUSES                                     | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | VFD                                                   | COMMON SDU                  | FUSES per circuit                         | UNIT MOUNTED                                 | UNIT MOUNTED                                            |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 5      | KAS                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | STAR DELTA                                            | MCCB                        |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFT STARTER                                          | SDU                         | FUSES                                     | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | VFD                                                   |                             |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | STAR DELTA                                            | MCCB per circuit            |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFT STARTER                                          | COMMON SDU                  | FUSES per circuit                         | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | VFD                                                   |                             |                                           | UNIT MOUNTED SINGLE PANEL (*Refer Note No.2) |                                                         |  |
|        | TRIPLE              | STAR DELTA                                            | MCCB per circuit            |                                           | UNIT MOUNTED 2 PANEL                         |                                                         |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 6      | KAA                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | STAR DELTA                                            | MCCB                        |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFTSTARTER                                           | SDU                         | FUSES                                     | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | VFD                                                   |                             |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | STAR DELTA                                            | MCCB per circuit            |                                           | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | SOFTSTARTER                                           | COMMON SDU                  | FUSES per circuit                         | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     | VFD                                                   |                             |                                           | UNIT MOUNTED SINGLE PANEL (*Refer Note No.2) |                                                         |  |
| TRIPLE | STAR DELTA          | MCCB per circuit                                      |                             |                                           | UNIT MOUNTED 2 PANEL                         |                                                         |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 7      | KCM                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | INVERTER (inbuilt in compressor)                      | SDU with door handle        | FUSES per circuit (Fuses not required for | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | DOUBLE CKT          | INVERTERS (inbuilt in compressor)                     | COMMON SDU with door handle | FUSES per circuit                         | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | TRIPLE CKT          | INVERTERS (inbuilt in compressor)                     | COMMON SDU with door handle | FUSES per circuit                         | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 8      | KAC                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | 11 TR               | DOL                                                   | NA                          | MPCB per compressor                       | UNIT MOUNTED SINGLE PANEL                    |                                                         |  |
|        | 22 TR               |                                                       |                             |                                           |                                              |                                                         |  |
|        | 33 TR               |                                                       |                             |                                           |                                              |                                                         |  |
|        | 44 TR               |                                                       |                             |                                           |                                              |                                                         |  |
|        |                     |                                                       |                             |                                           |                                              |                                                         |  |
| 9      | KSM                 |                                                       |                             |                                           |                                              |                                                         |  |
|        | SINGLE CKT          | VFD                                                   | ACB/MCCB                    |                                           | UNIT MOUNTED                                 | UNIT MOUNTED STARTER PANEL & FREESTANDING INCOMER PANEL |  |

|                                                                                                                                                                           |                                                                                                                                                                                |                  |                  |
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| Document Name: STANDARD OFFERINGS FOR ELECTRICAL PANEL.                                                                                                                   |                                                                                                                                                                                |                  |                  |
| Document No:                                                                                                                                                              | ESP-17-18-101                                                                                                                                                                  | Revision: R04    | Date: 14/04/2022 |
| Prepared By: MRM                                                                                                                                                          | Revised By: SRS                                                                                                                                                                | Approved By: ASP | Sheet: 2 of 2    |
| Common points applicable to all models                                                                                                                                    |                                                                                                                                                                                |                  |                  |
| 1                                                                                                                                                                         | Above mention MCCB is with Shortcircuit protection (50KA).                                                                                                                     |                  |                  |
| 2                                                                                                                                                                         | For KSC & KCM (Tripple Ckt) Series cable entry is from TOP & For Scroll chillers (KAC011 & KAC022 )cable entry is from left side .For other series cable entry is from BOTTOM. |                  |                  |
| 3                                                                                                                                                                         | Control & power cables are Copper flexible type.                                                                                                                               |                  |                  |
| 4                                                                                                                                                                         | For any standalone panel, entire interfacing cabling is in customer scope. Interfacing drawing will be submitted to customer .                                                 |                  |                  |
| 5                                                                                                                                                                         | Control and starter panel boxes are of IP54 protection class.                                                                                                                  |                  |                  |
| 6                                                                                                                                                                         | Control and starter panel boxes are of IP55 protection class for KAS/KAA chiller .                                                                                             |                  |                  |
| Accessories /Options with additional cost and without additional lead time (Subject to material availability)                                                             |                                                                                                                                                                                |                  |                  |
| 1                                                                                                                                                                         | ENERGYMETER/MFM                                                                                                                                                                |                  |                  |
| 2                                                                                                                                                                         | HOOTER                                                                                                                                                                         |                  |                  |
| 3                                                                                                                                                                         | Phase indicating lamp.                                                                                                                                                         |                  |                  |
| 4                                                                                                                                                                         | Voltmeter                                                                                                                                                                      |                  |                  |
| 5                                                                                                                                                                         | Ammeter                                                                                                                                                                        |                  |                  |
| 6                                                                                                                                                                         | MCCB with Earthfault protection,microprocessor based trip                                                                                                                      |                  |                  |
| Accessories /Options with additional cost and with additional lead time                                                                                                   |                                                                                                                                                                                |                  |                  |
| 1                                                                                                                                                                         | 3 Phase 3 wire system.                                                                                                                                                         |                  |                  |
| 2                                                                                                                                                                         | 110V control supply.                                                                                                                                                           |                  |                  |
| 3                                                                                                                                                                         | LOTO arrangement.                                                                                                                                                              |                  |                  |
| 4                                                                                                                                                                         | Standalone Harmonic Filter panel (*Refer Note No.3)                                                                                                                            |                  |                  |
| 5                                                                                                                                                                         | Purging panel for corrosive environments.                                                                                                                                      |                  |                  |
| 6                                                                                                                                                                         | Space heater ,Thermostat & Tubelight                                                                                                                                           |                  |                  |
| NOTES:                                                                                                                                                                    |                                                                                                                                                                                |                  |                  |
| 1. Above are standard offerings ,any other requirements from this should be confirmed from engg for feasibility ,additional cost and additional lead time as applicable . |                                                                                                                                                                                |                  |                  |
| 2. For Higher rating chiller models ,Consult with engineering team for VFD panel size & mounting .                                                                        |                                                                                                                                                                                |                  |                  |
| 3 . AHF selection shall be taken from engineering . Cabling shall be in customer scope for AHF standalone panel .                                                         |                                                                                                                                                                                |                  |                  |